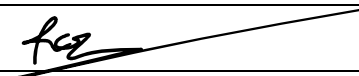


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Assignment, Report & Laboratory Coversheet for Individual & Group Assignment

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UNIT NAME Engineering Research Project Part 1		UNIT CODE GENG4411
TITLE/TOPIC OF ASSIGNMENT Progress Report		NAME OF LECTURER/TUTOR Kieran Scott Quirke-Brown
DATE/TIME DUE 19 th August 2026 11:59pm		DATE/TIME SUBMITTED 19 th August 2026 11:45am
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GENG4411: Progress Report

Detect, Then Verify: A Confidence Gated Detector to VLM Pipeline for Real Time Occluded Person Search and Rescue

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Project Type: Design and Build

Date: 19th August 2026

Nomenclature

fps	Frames Per Second
GPU	Graphics Processing Unit
HERIDAL	Croatian Center for Robotics aerial imagery dataset for SAR person detection (Božić-Štulić et al.)
HPI	Human Presence Indicator (e.g. backpacks, clothing, shelters, footprints)
mAP@50	mean Average Precision at an IoU threshold of 0.50
NVMe SSD	Non-Volatile Memory express Solid State Drive
PX4	Open-source flight control software/autopilot
R-CNN	Region-based Convolutional Neural Network
RGB	Red, Green, Blue (standard colour imagery)
ROS2	Robot Operating System 2
SAR	Search and Rescue
SITL	Software-In-The-Loop (simulation)
TensorRT	NVIDIA inference optimisation library/runtime
UAV	Unmanned Aerial Vehicle
VLM	Vision-Language Model
YOLO	You Only Look Once (real-time object detection model family)

1 Problem Statement

Search and Rescue operations in low visibility, dense terrain are hampered by poor sight lines and vegetative occlusion, degrading both human search effectiveness and existing vision based detection architecture. Prior UAV based human detection work for SAR has good performance in sparsely vegetated terrains. The Božić-Štulić et al.'s HERIDAL system reaches a detection rate of 88.9% at a 45 to 60 m altitude [1], although the subjects in the imagery are visible clearly from the height of the UAV. Detectors trained on unoccluded and standard pedestrian data will largely fail on partially occluded and non standing subjects (humans) under tree foliage [2]. There is no study which has established how much decay in performance occurs under severe bush occlusion or what onboard architecture recovers best.

Detections have to run onboard in real time, on local hardware not a workstation GPU. An architecture benchmarked against HERIDAL is the Faster R-CNN, managed only 5 to 17 fps on a workstation GPU [3]. A YOLO model trained on HERIDAL via two stage transfer learning reached 0.802 mAP@50, although inference on a Jetson Nano ran only at around 2 fps without TensorRT or quantisation [4]. The Jetson Orin Nano Super used here has substantially higher compute and applying the optimisations makes real time embedded detector plausible, though unproven. What's open is whether it holds sufficient accuracy under occlusion and if a secondary, lightweight vision language model (VLM) can recover reliability where a single pass detector fails. This is done by classifying high confidence detections into categories (clothing, shelter, equipment etc.), while low confidence detections are verified as true or false.

Research Hypothesis: A low altitude (ideally 5 to 10 m) UAV running an onboard, quantised detection pipeline, operating deliberately at a reduced confidence threshold with high confidence detections routed to a lightweight VLM for HPI classification and low confidence detections for verification, will recover occluded real positives that a detector only pipeline can discard, improving on mAP@50 and recall on occluded bush images without precision net loss and also sustaining a real time inference on a Jetson Orin Nano Super. Secondly, sharing these detections across a multi drone fleet will reduce time to full area coverage and redundant searching relative to non collaborative, independent work.

2 Objectives

- **O1:** By how much do precision, recall, and mAP@50 drop when a baseline detector trained on open terrain SAR data [1] is applied to occluded bush imagery and freshly collected RGB bush imagery set?
- **O2:** Among candidate detector architectures, including the YOLO-family and state-space/Mamba-hybrid models [4, 5], which offers the best balance of accuracy, latency, and resource usage on occluded aerial imagery on the Orin Nano Super?
- **O3:** Can HPIs, including backpacks, discarded clothing, shelters plus additional classes explored during dataset construction be detected reliably enough that a human reviewer can act on alert, remaining operationally useful when the person themselves is fully occluded?
- **O4:** Does routing low confidence detections to an onboard VLM for verification, and high confidence detections for indicator classification, recover recall and mAP@50 without a net precision loss or without breaching the real-time latency budget on the Orin Nano Super?
- **O5:** Among candidate lightweight VLMs suited to onboard deployment [6], which one best balances verification and classification accuracy against inference latency on the Orin Nano Super?
- **O6:** Does real time detection sharing across a multi drone fleet reduce time to full coverage and redundant area search, compared to independent, non collaborative search.

3 Methodology

3.1 Literature Review: Research behind SAR detection follows the HERIDAL lineage of datasets [1], using the YOLO-family and state-space/Mamba hybrid detectors [5], targeting occlusion directly [2].

RGB-thermal fusion [7] and onboard VLMs on Jetson hardware [6] were reviewed, with the thermal fusion scoped out based on cost limitations.

3.2 Construction of Dataset: Training will mainly draw on HERIDAL [1], extended through synthetic occlusion overlays and also new UAV imagery over bush terrain. HPI classes such as backpacks, clothing, footprints, shelters, etc. require a brand new annotation since no data set specifically covers them, following the two stage transfer learning protocol used for the detector models [4].

3.3 Detector Development and Training: Chosen detectors for evaluation [4, 5] will be trained on identical data, then optimised for local embedded inference on the Jetson Orin Nano Super and Arducam AR0822 camera. Detections with low confidence will be routed to the best VLM identified in **O5** for verification, whereas the high confidence detections are sent to the VLM for indicator classification, with the confidence threshold for routing set through validation set sweep to maximise recovered recall subject to precision without excessive VLM calls.

3.4 Evaluation: Conducted Offline evaluation will be used to report precision, recall and mAP@50 on an occluded test set. Separately, on-device, local evaluation will measure frame rate, latency and power usage/draw on the Jetson Orin Nano Super, distinguishing inference time of the model from end to end pipeline latency [6]. The VLM stage's contribution and usefulness will be evaluated using an ablation.

3.5 Simulation and Field Evaluation: A Gazebo Harmonic and PX4 SITL simulation with a height-map derived bush land and suburban world with the addition of multi drone support is already operational. Fleet coordination is to be evaluated in simulation by comparing the time taken for full coverage and redundant area re-search between independent and collaborative search plans (**O6**), with later integration of communication infrastructure from a triangulation based project run in parallel. Physical flight testing is dependent on build completion with simulation and bench testing as a fallback.

3.6 Challenges and Mitigation: The primary risk is that no occlusion based bush dataset exists, which is addressed through synthetic occlusion augmentation combined with transfer learning from HERIDAL [1]. Latency of the onboard VLM could breach the real time compute power/budget. This is mitigated by triggering it only on low confidence detections and with the use of a quantised model with sub 4B parameters. The Jetson's 8 gigabyte unified memory cap is managed via quantisation, NVMe SSD swap and a benchmarking step before implementing a model size. Training dataset drawn from HERIDAL was collected at 45 to 60 m altitude [1], whereas this project targets 5 to 10 m. The resulting viewpoint and scale difference will be addressed alongside occlusion augmentation during dataset collection.

4 Expected Outcomes

It is expected that detection performance will fall below the HERIDAL's 88.9% rate under open terrain [1], because detectors trained on unoccluded datasets already fail under occlusion [2]. The occlusion augmented, transfer learned detector is expected to have a meaningful recovery rate of that gap in mAP@50, informed by the 0.802 which has been reached by similar spec embedded detectors on less occluded HERIDAL datasets [4], while sustaining a real time inference on the Jetson Orin Nano Super. Two outcomes that remain uncertain. One is whether the VLM verification stage recovers reasonable accuracy to justify the added end to end latency. Secondly, whether HPI classes can be reliably learned from newly annotated datasets with limited examples is also equally unresolved. There is still weight to a negative result. If detection via RGB only can't recover meaningfully under occlusion, there is direct evidence that multi modal sensing like RGB-thermal fusion [7] is needed, not just optional for heavy occlusion SAR. The significance of this project lies in furthering UAV search into areas where current vision systems fail and also to produce an occlusion focused benchmark, comparison of an embedded architecture and a HPI dataset that doesn't exist yet in literature.

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Figure 1: Gantt Chart showing milestones and tasks from Thesis Semester 1 (left) and Thesis Semester 2 (right).

